



Dynamic planning of crowd evacuation path for metro station based on Dynamic Avoid Smoke A-Star algorithm

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ABSTRACT

Considerable casualties can easily be caused once a fire occurs in a metro station, and a safe and effective evacuation path in time should be provided, taking on critical significance in the rapid evacuation of the crowd. In existing research, crowd evacuation paths have been generally planned without considering the fire environment and the real-time effect of fire products on the evacuation paths, such that the planned paths do not fit the realistic environment of crowd evacuation. An algorithm is proposed in this study to dynamically plan evacuation paths in fire scenarios in metro stations. First, the equivalent length of the path under the effect of fire on evacuation speed is determined in accordance with the effects of ambient temperature, visibility, and CO concentration, and then the fire risk model is built. Second, the fire risk model is incorporated into the A-Star algorithm, and the evaluation function of the A-Star algorithm is optimized, such that the determined path is capable of avoiding areas with higher fire risk. Subsequently, a dynamic update mechanism considering time factor is introduced to update the search environment information data of the algorithm in real time for dynamic path planning, with the aim of coping with the dynamically changing fire environment. Lastly, the A-Star algorithm is optimized, and the Dynamic Avoid-Smoke A-Star (DASA-Star) algorithm is built. As indicated by the simulation results, the DASA-Star algorithm is capable of making a trade-off between the fire risk and geometric length of the path in an underground station fire scenario and achieving dynamic planning of evacuation paths based on FDS fire simulation results. Accordingly, the algorithm conforms to the real-time requirements of path planning under fire conditions, and it is capable of more effectively planning the optimal evacuation path under the effect of fire.

1. Introduction

The design of metro station is dependent on several factors (e.g., topography and development level), with deep underground burials covered typically. Metro stations, i.e., a special public building situated deep underground, exhibit high population density, large pedestrian flow, as well as complex internal structures (Li et al., 2016; Zhou et al., 2021). There exist a considerable number of uncertainty factors for the evacuation of people from metro station in emergency event, which comprise limited time for safety evacuation, difficulty in selecting effective evacuation paths, and so forth, thus easily triggering significant casualties (Chen et al., 2017; Huang et al., 2022; Li et al., 2020a; Song et al., 2013). Although emergency evacuation facilities and strategies (e.g., safety signs and evacuation plans) are pre-designed for metro stations

according to building requirements, they provide limited practical assistance for evacuees during emergencies due to uncertain dynamic changes of evacuation environmental (Lo, 1996; Pu and Zlatanova, 2005). For instance, the spread and concentration of flames and smoke in fire are capable of reducing or eliminating the safety of several evacuation paths, delaying evacuation and causing fatal consequences for stranded individuals (Choi and Chi, 2019; Ran et al., 2014). People in emergency situations lose their ability to adapt to their surroundings, such that sufficient knowledge regarding their surroundings is difficult to acquire in a short period, with their behavior exhibiting herd instinct and familiarization tendencies (Helbing et al., 2000). As Revealed by the above-mentioned phenomena, the fundamental reason why pedestrian evacuation efficiency and safety are affected is that pedestrians cannot obtain adequate information regarding the evacuation environment and

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fully exploit environmental resources. Measuring and analyzing factors for the safety of pedestrian evacuation based on metro station environments and emergency incident information, using relevant path planning algorithms to achieve the dynamic programming of the optimal pedestrian egress paths and promptly provide scientific and reasonable evacuation paths to people in the station, such that the above-mentioned issues can be more effectively resolved (Cuesta et al., 2017; Galea et al., 2017; Samah et al., 2015; Zhang et al., 2022a).

Fire is accompanied by a large number of toxic and harmful smoke. Moreover, the relatively closed station space and the “piston effect” of train operation are capable of exacerbating the difficulty of smoke emission, while the diffusion and gathering of smoke and the vertical channel of the “chimney effect” significantly hinder the crowd evacuation in the station (Cai et al., 2022; Seike et al., 2016; Seike et al., 2021). The fire smoke can notably stimulate the evacuees in the metro station. Consequently, panic psychology will be triggered, the visibility of the evacuees will be affected, and their vision in the evacuation process will be greatly reduced. Besides, the toxic and harmful gases and the thermal radiation effect of the smoke pose threaten the occupant safety in the metro station (Coskun et al., 2022; Li et al., 2020b). Accordingly, the shortest geometric length of the path is not necessarily the only criterion for evacuation path planning to ensure the safety and efficient evacuation in fire (Zhang et al., 2019).

In general, the crowd evacuation has a strong purpose, and its evacuation strategy is not static, whereas it is constantly adjusted with the change of environmental state. On that basis, the current evacuation environment state information should be used, and the future evacuation environment state information should be considered (Shi et al., 2009). In the metro station fire environment, the irregular spread of fire and the diffusion and gathering of fire smoke are constantly changing with time, and the static environmental information based on a certain moment is unable to meet the practical evacuation requirements. In addition, the behavioral actions of the crowd in the evacuation process are significantly complex, such that the evacuation path planning is triggered with certain specificity and complexity (Han et al., 2017), such that the evacuation path planning for metro station fire conditions is different from the conventional path planning. In time-sensitive fire emergencies, the occupant safety evacuation may benefit from the combination of fire risk modeling and path planning algorithms (Huang et al., 2019).

Path planning algorithm is one of the key technologies of computer evacuation simulation, which has been widely used in the field of evacuation. Evacuation path planning is not consistent with the general path planning problem, which should consider the static environment constraints and solve the dynamic environment information change. Moreover, it refers to a multi-constrained dynamic path planning problem. Choi and Chi, (2019) predicted the safety state of nodes in a building based on the A-Star algorithm using a fire dynamics simulator to simulate smoke dispersion to plan safer evacuation paths. Liu et al., (2016) updated the ant colony pheromone through quantum representation using Ant Colony Optimization algorithm to avoid obstacles more effectively and increase the planning path safety. To enhance the realism of evacuation path planning, some scholars have considered the location state of pedestrians in practical situations following known environmental information and introduced information (e.g., the number of people and initial location) in the algorithm. Wong et al., (2015) divided the initial crowd by number for path planning using the Particle Swarm Optimization algorithm to increase the efficiency of crowd evacuation.

The goal of evacuation path planning is to find the optimal path under the current situation. Its accuracy depends on the accuracy of the environmental information obtained. It is necessary to consider the unknown information that changes in real time, and it is more demanding and more complex to obtain a more realistic evacuation demand path planning result (Peng et al., 2019). The environmental information varies dynamically under the effect of unexpected events, which brings the requirements (i.e., real-time and robustness) to

evacuation path planning. Xu et al., (2022) changed the heuristic function of ant colony algorithm and the update strategy of pheromone based on the results of PyroSim fire simulation to achieve real-time evacuation path planning under fire conditions. Zhao et al., (2018) developed a path planning method based on parallel architecture asynchronous reinforcement learning algorithm for path planning, which dynamically changes the action strategy in accordance with the environment to address the problem of dynamic path planning for occupant evacuation. Furthermore, some scholars have optimized the path algorithm using the method of real-time update of environmental information to tackle down the problem of real-time change of environmental information (e.g., updating the environmental information data with a certain frequency and then periodically modifying the planned path) (He et al., 2022). Khakzad, (2023) considers the threat of tank fire, uses Dijkstra algorithm to plan the evacuation path, and combines mathematical programming to allocate the crowd reasonably to reduce the risk of casualties. He et al., (2023) based on the Dynamic A-Star algorithm, considering the dynamic, multi-hazard and domino effect of fire, explosion and toxic substance release accidents, proposed an evacuation path planning method suitable for multi-hazard accidents in the chemical industry. Huang et al., (2021) established a fire database, combined with human tolerance for dynamic reconstruction, and used the A-star algorithm to plan the evacuation path of the reconstructed scene, which realized the dynamic planning of the evacuation path. Gibson et al., (2020) proposed a dynamic path planning algorithm based on the A-Star algorithm. This algorithm periodically updates the generated paths using state feedback from the real environment; it shows the advantage of a simpler structure and easy implementation, while it is not determined by other algorithms. Shen et al., (2015) determined the diffusion concentration of chlorine in a chemical park using a Gaussian plume model and incorporated Dijkstra algorithm in order to calculate the optimal evacuation path to an emergency shelter with the minimum toxic dose.

As revealed by the existing research and analysis of the problems regarding evacuation path planning, using path dynamic planning can effectively enhance its security and real-time performance (Kim et al., 2023), whereas some difficulties and shortcomings remain in the research of path dynamic planning algorithms. Although several algorithms have been extensively employed in sparse path dynamic planning with the advantages of high efficiency and intelligence, their path planning results may still fall into local optimum, whereas the A-star algorithm in accordance with the heuristic principle is not correlated with this problem (Zeng and Church, 2009). A-Star algorithm can be considered the optimization direction of the Dijkstra algorithm, based on which heuristic function is introduced, which shows the advantage of accurate and fast compared with Dijkstra algorithm (Hart et al., 1968; Persson and Sharf, 2014; Xu et al., 2007). As an efficient path search algorithm, A-star algorithm is widely used in a variety of evacuation scenarios. It solves the evacuation path planning problem by calculating the shortest path from the starting point to the end point. In the actual evacuation scenario, evacuation path planning should not only consider the path length, but also consider other factors, such as evacuation time, safety, and congestion. Aiming at the special needs of evacuation path planning, the traditional A-star algorithm is improved, such as introducing machine learning technology to predict the flow dynamics in the evacuation process, or improving the heuristic function to improve the efficiency of path search. With the advancement of technology and the in-depth understanding of evacuation requirements, the A-star algorithm will continue to be optimized and improved, including the improvement of algorithm efficiency, the enhancement of adaptability to complex environments, and the integration with other algorithms to solve more complex evacuation path planning problems.

A dynamic planning model of evacuation path considering the effect of real-time changes of fire is built in this study based on the optimized A-star algorithm for the metro station scenario. Considering the influence of temperature, visibility and CO concentration on evacuation

speed, a fire risk model is established, which is coupled with the algorithm to change the evaluation function of the algorithm, so that it can calculate a safer evacuation path and meet the requirements of safe evacuation. The introduction of time dynamic update mechanism has the ability to cope with the dynamic changes of environmental information. It is different from the reconstruction planning based on the static information at a certain time after updating the map data. The DASA-Star algorithm considers the processing of environmental information changes in the search process. With the help of fire smoke data simulated by FDS, the algorithm searches for environmental information data according to the real-time update algorithm of personnel movement time, continuously optimizes the path, and fits the real evacuation process to achieve the dynamic planning of the best escape route. The dynamic planning algorithm model of evacuation path established in this study can be used for evacuation path planning in subway station fire scene, and provide decision support for emergency response of subway station.

Fig. 1 presents the technology roadmap of this study. The rest of this study is organized as follows. In Section 2, the metro station fire risk model is built. In Section 3, the Dynamic Avoid-Smoke A-Star algorithm is presented, which is termed the DASA-Star algorithm. In Section 4, the algorithm simulation experiments are performed with the results of the metro station fire simulation. In Section 5, the effectiveness of the DASA-Star algorithm is analyzed and discussed. In Section 6, the conclusion and prospect are drawn. Lastly, in Section 7, the limitations of this study are analyzed.

2. Fire risk modeling for metro stations

The search for the shortest path and the pursuit of the shortest evacuation time in a metro station fire scenario do not conform to the requirements for safe evacuation of people, the safety of evacuation paths takes on greater significance in a fire scenario. Thus, a relevant fire risk model should be built to represent the effect of fire and its products

on the safety of evacuation paths, where the core refers to identifying the fire products in the environment and quantifying the effect of fire and its products on evacuation.

2.1. Fire risk model

In practical fire scenarios, besides the high temperatures generated by the fire, a wide variety of fire products can affect occupant evacuation to a certain extent (e.g., physiological and psychological aspects) (Fang et al., 2012; Lo and Fang, 2000). Drawing on existing research on the effects of fire products on evacuation, three factors of temperature, visibility and carbon monoxide concentration in the evacuation environment are considered and the effect on evacuation speed is analyzed. Moreover, the temperature effect function $R_1(T)$, visibility effect function $R_2(K_c)$, and CO concentration effect function $R_3(\rho_{CO})$ are introduced to represent the correlation between the three factors and evacuation speed (Xu et al., 2022). Furthermore, the fire smoke effect function f is introduced to represent the combined effect of fire product evacuation paths. f is determined by the temperature effect function $R_1(T)$, visibility effect function $R_2(K_c)$, and CO concentration effect function $R_3(\rho_{CO})$.

$$f(T, Kc, \rho_{\text{co}}) = R_1(T) \times R_2(Kc) \times R_3(\rho_{\text{co}}) \quad (1)$$

Accordingly, the effective evacuation speed v_t under the effect of fire is expressed as:

$$\boldsymbol{v}_t = \boldsymbol{v}_0 \times f(T, K_c, \rho_{\text{co}}) = \boldsymbol{v}_0 \times R_1(T) \times R_2(K_c) \times R_3(\rho_{\text{co}}) \quad (2)$$

where v_0 denotes the movement speed of evacuees under normal conditions, v_t represents the equivalent evacuation speed of evacuees under the effect of fire.

Given the analysis of the effect of the parameters correlated with the fire products on the evacuation speed in fire scenario, the equivalent length L of the evacuation path under the effect of fire is expressed as:

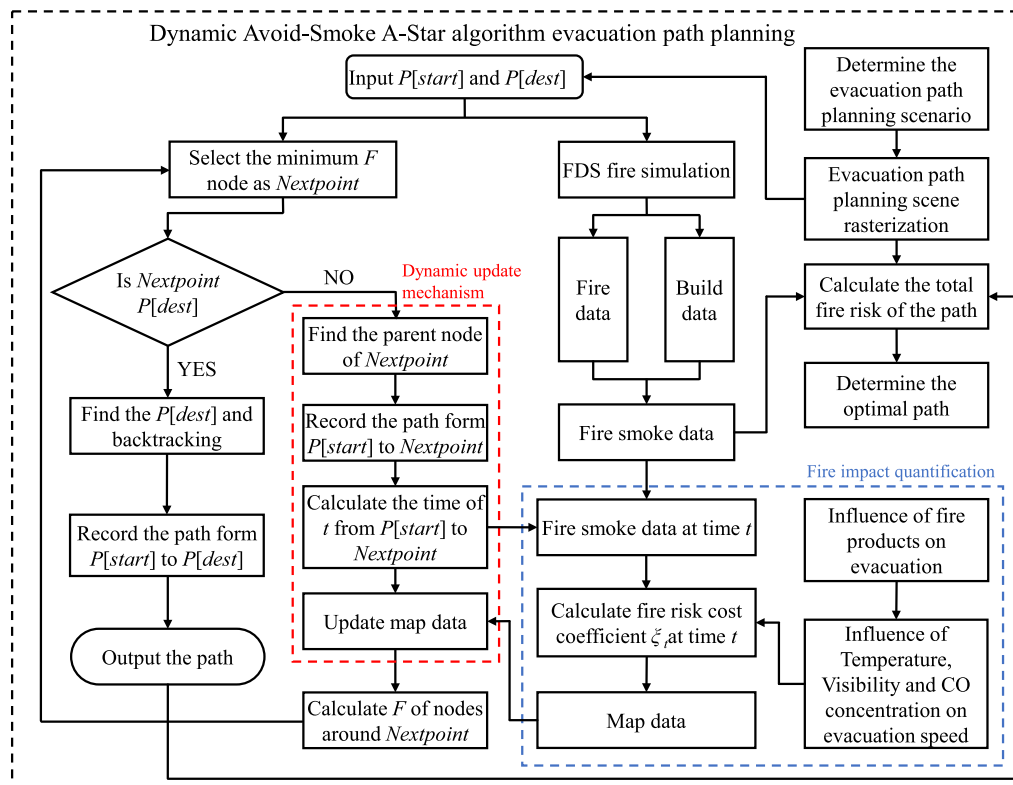


Fig. 1. Flow chart for Dynamic Avoid-Smoke A-star algorithm.

$$L = \frac{l}{v_t} \times v_0 \quad (3)$$

where l represents the practical geometric length of the evacuation path, while L represents the effective length of the evacuation path under the effect of a fire.

Under the effect of a fire, the speed of passage on different paths is different, resulting in different equivalent lengths of paths in the fire environment, hence this study proposes to express this effect by the fire risk cost factor ξ , whose expression is as follows:

$$\xi = \frac{L}{l} = \frac{1}{f(T, K_c, \rho_{co})} \quad (4)$$

According to Eq. (4), it can be seen that the equivalent length of evacuation paths is proportional to the fire risk cost coefficient ξ . When the practical geometric lengths of different evacuation paths are the same, the larger the fire risk cost coefficient ξ and the longer the equivalent length L of evacuation paths under the effect of fire. The higher the degree of effect of fire and its products on evacuation paths, the greater the fire risk. Thus, the fire risk cost coefficient ξ can be used to represent the fire risk of evacuation path in fire scenario. To better map the fire risk in a metro station evacuation scene, the fire risk cost coefficient ξ is normalized by the extreme difference.

It is worth noting that if any position in the evacuation scene meets any of the following conditions: (1) temperature greater than 120 °C, (2) CO concentration greater than 0.1 %, (3) visibility less than 3 m, it is considered that the fire risk at that position has reached the critical value of safety evacuation (Xu et al., 2022), and it cannot be used for evacuation.

2.2. Temperature effect function

Temperature exerts different effects on the evacuation speed of personnel at different stages. Besides, the initial stage of the fire exerts a positive incentive effect on the evacuation speed of personnel at a low temperature. Nevertheless, as the smoke accumulates and the ambient temperature is elevated to a certain value, the evacuation speed of personnel will be adversely affected. When the ambient temperature reaches a critical value of 120 °C, the complete loss of movement of personnel will be triggered. In accordance with the equation proposed by Milke, (2000), the function of temperature effect on evacuation speed $R_1(T)$ is expressed as follows:

$$R_1(T) = \begin{cases} 1 & T_0 < T_s \leq T_{c1} \\ \frac{(v_{\max} - 1.2) \left(\frac{T_s - T_{c1}}{T_{c2} - T_{c1}} \right)^2}{v_0} + 1 & T_{c1} < T_s \leq T_{c2} \\ \frac{v_{\max}}{1.2} \left[1 - \left(\frac{T_s - T_{c2}}{T_d - T_{c2}} \right)^2 \right] & T_{c2} < T_s \leq T_d \end{cases} \quad (5)$$

where T_0 denotes the normal environmental temperature (°C), T_s represents the practical temperature in a fire environment (°C), v_0 expresses the normal walking speed of humans (m/s), which is taken as 1.2 m/s, $v_{\max}$ is the maximum walking speed of humans (m/s), which is taken as 5 m/s, T_{c1} is the temperature at which humans feel uncomfortable, which is taken as 30 °C, T_{c2} is the temperature at which humans are injured, which is taken as 60 °C, T_d is the temperature at which humans die, which is taken as 120 °C (Xu et al., 2022).

2.3. Visibility effect function

Fire smoke comprises solid particles, liquid particles, toxic and hazardous chemicals, and the shading of smoke shows a direct correlation with the concentration of the above-mentioned substances. Based on previous experience, the extinction coefficient K_c is employed to assess the shading of fire smoke, and the value of K_c exerts a direct effect

on the visibility of personnel in the evacuation environment (Nguyen et al., 2013; Wang and Zheng, 2013). The larger the value of K_c , the lower the visibility of personnel will be, thus hindering the observation of the surrounding environment. As a result, the efficiency of evacuation is reduced. Referring to Fridolf et al., (2019), the function of visibility on evacuation speed $R_2(K_c)$ is expressed as follows:

$$R_2(K_c) = \begin{cases} 1 & K_c < 1.2 \\ 1 - 0.3025K_c & 0.35 \leq K_c < 1.2 \\ 0.25 & K_c \geq 1.2 \end{cases} \quad (6)$$

where K_c is the obscuration coefficient (1/m).

2.4. CO concentration effect function

Fire smoke covers considerable toxic gases that have not been completely burned. Since the metro station is in underground and in a confined space, it is difficult to vent the toxic gases, thus significantly affecting evacuation. Moreover, CO is the greatest threat to the occupant safety evacuation. Milke (2000) experimentally concluded that the effect of CO on evacuation is primarily in the speed of evacuation of people. With reference to the study of Milke (2000), the effect function of CO concentration on evacuation speed $R_3(\rho_{co})$ is expressed as:

$$R_3(\rho_{co}) = \begin{cases} 1 & \rho_{co} < 0.1 \\ 1 - (0.2123 + 1.788\rho t) & 0.1 \leq \rho_{co} < 0.25 \\ 0 & \rho_{co} \geq 0.25 \end{cases} \quad (7)$$

where ρ_{co} denotes the volume concentration of CO (%); t represents the time of personnel exposure to CO (s)

3. Improved A-Star algorithm

In this section, an algorithm for dynamic planning of evacuation paths, i.e., Dynamic Avoid-Smoke A-Star algorithm, is proposed based on the traditional A-Star algorithm for improvement, and its principles are introduced. First, the traditional A-Star algorithm is introduced and the shortest path problem is analyzed. Second, the fire risk model is substituted into the A-Star algorithm to change the decision principle of the algorithm. Lastly, the Dynamic Avoid-Smoke A-Star algorithm is built by introducing a dynamic update mechanism to cope with the dynamic changes of fire environment information.

3.1. Traditional A-Star algorithm

The A-Star algorithm was first proposed and described in detail by (Hart et al., 1968), aims to search for the least costly (e.g., shortest distance, shortest time, etc.) path from the starting point $P[start]$ to the end point $P[dest]$. In the actual evacuation path planning, the A-star algorithm has been widely used. For example, in scenarios such as internal evacuation of buildings and urban traffic network evacuation, the A-star algorithm can be used to efficiently and quickly plan a safe and effective evacuation path. The A-star algorithm can also be used to simulate and optimize the evacuation process and evaluate the effectiveness of different evacuation strategies. The traditional A-Star algorithm pseudocode is shown in Table 1.

where $P[i]$ and $P_j[j]$ denote the current node and the child nodes around the current node $P[i]$, respectively, and the nodes in the open and closed lists store the reachable and unreachable nodes, respectively. In the traditional A-Star algorithm, the evaluation function $F(P[i])$ of $P[i]$ is defined as:

$$F(P[i]) = G(P[i]) + H(P[i]) \quad (8)$$

where $G(P[i])$ denotes the practical cost function from the starting point $P[start]$ to the current node $P[i]$, $H(P[i])$ denotes the predicted cost function from the current node $P[i]$ to the end point $P[dest]$ (Bagheri et al., 2021). The predicted cost of the current node $P[i]$ to the end point

Table 1
Pseudocode of traditional A-Star algorithm.

Algorithm 1: Traditional A-Star algorithm ($P[start] \rightarrow P[dest]$)	
1:	Mark $P[start]$ as <i>openlist</i> .
2:	while <i>openlist</i> $\neq$ empty do
3:	Select the node $P[i]$ from the <i>openlist</i> whose value of evaluation function $F(P[i])$ is smallest.
4:	if $P[i] = P[dest]$ then
5:	return "path is found".
6:	else
7:	Mark $P[i]$ as <i>closelist</i> .
8:	if Successor $P_i[j]$ of node $P[i]$ does not belong to <i>closelist</i> or <i>obstacle</i> node then
9:	Mark $P_i[j]$ as <i>openlist</i> and calculate $F(P_i[j])$.
10:	if $P_i[j]$ belongs to <i>openlist</i> and $F(P_i[j])$ is smaller than $F(P_m[j])$ when $P[m]$ was marked as <i>closelist</i> then
11:	Set parent node of $P[j]$ as $P[i]$, and $F(P[i]) = F(P_i[j])$.
12:	end if
13:	end if
14:	end if
15:	end while
16:	return "the path cannot be found".

$P[dest]$ is determined by Euclidean distance, Manhattan distance, Diagonal distance, etc. (Liang et al., 2021; Soltani et al., 2002). In the two-dimensional data calculation of sparse path planning, the Euclidean distance can show the distance between two points intuitively. Accordingly, the Euclidean calculation method is used in this study.

The traditional A-star algorithm only takes the shortest path geometric length as the only criterion in the evacuation path planning process, regardless of the risk factors caused by fires, and its path planning results are shown in Fig. 2. The paths of the nodes (3,3) to (3,4) stages are close to the fire source, resulting in a higher fire risk and lower safety for the paths at this stage.

3.2. A-star algorithm considering fire risk

In the evacuation path planning of a metro station, the safety of the path is of more concern than the geometric length of the path. When the fire reaches a certain level, not only the flame burning area will hinder the safe evacuation of people, but also the fire smoke and other gases will threaten the safety of people during evacuation. This section couples the A-Star algorithm with the fire risk model built in Section 2, and takes the equivalent length of the path under the effect of fire as the new decision principle of the algorithm, so that the algorithm takes into account the effect of fire and its products on evacuation in path planning, with a view to planning an evacuation path that can avoid the area where the fire has a negative impact.

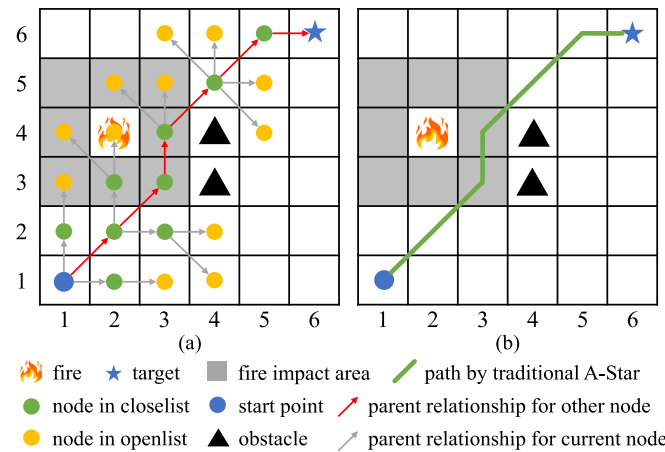


Fig. 2. The traditional A-Star algorithm path planning grid graph, (a) shows the application of related nodes in path planning, the arrow represents the parent-child relationship from the parent node to the child node, (b) the traditional A-Star algorithm path planning results.

In the A-Star algorithm with coupled fire risk, the shortest practical geometric length of the evacuation path is no longer taken as the decision principle in path planning, whereas the path is selected based on the equivalent length of the evacuation path under the effect of fire. Accordingly, the practical cost function in the evaluation function (Eq. (8)) is optimized, and the fire risk cost coefficient ξ is substituted into the practical cost function $G(P[i])$. Next the new $G(P[i])$ is determined as Eq. (9):

$$G(P[i]) = \sum_{n=1}^k l(n)\xi(n) = \sum_{n=1}^k L(n) \quad (9)$$

where k denotes the number of nodes from $P[i]$ to $P[start]$, i.e. $P[i]$, the parent of $P[i]$, the parent of the parent point of $P[i]$, ... $P[start]$; $L(n)$ represents the equivalent length from the $n-1$ node to the n node under the effect of fire; $\xi(n)$ expresses the fire risk cost coefficient at the node $P[n]$; $l(n)$ denotes the practical geometric length from the $n-1$ th node to the n th node (Euclidean distance); n represents the sequence of nodes starting from the starting node $P[start]$.

Where $l(n)$ is presented in Eq. (10), and it is assumed that the coordinate of the n node is (x_n, y_n) .

$$l(n) = \sqrt{(x_n - x_{n-1})^2 + (y_n - y_{n-1})^2} \quad (10)$$

The predicted cost function $H(P[i])$ represents the prediction of the shortest path cost from the current node $P[i]$ to the end point $P[dest]$. It is assumed that the coordinates of $P[i]$ are denoted as $(x_{P[i]}, y_{P[i]})$ and the coordinates of $P[dest]$ are expressed as $(x_{P[dest]}, y_{P[dest]})$, such that the predicted cost function $H(P[i])$ is expressed as:

$$H(P[i]) = \sqrt{(x_{P[dest]} - x_{P[i]})^2 + (y_{P[dest]} - y_{P[i]})^2} \quad (11)$$

Thus, the evaluation function $F(P[i])$ of the optimized A-star algorithm coupled with fire risk is:

$$F(P[i]) = \sum_{n=1}^k l(n)\xi(n) + H(P[i]) \quad (12)$$

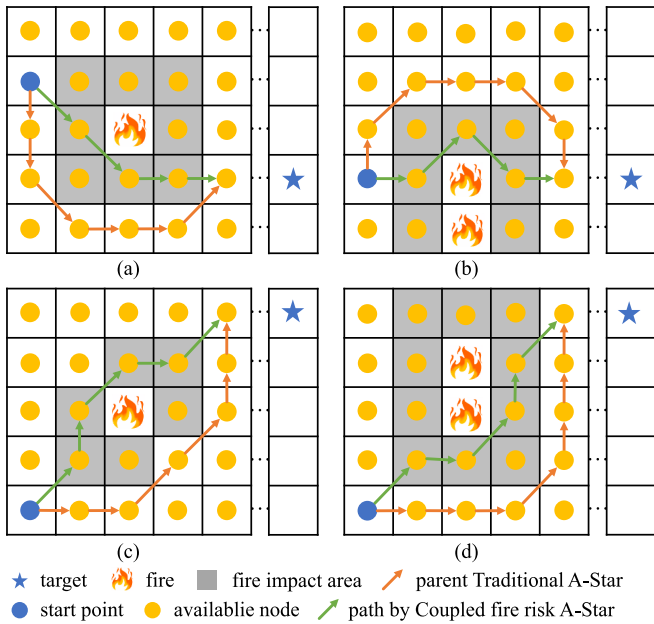
Based on the traditional A-Star algorithm, the pseudocode of the coupled fire risk A-Star algorithm is shown in Table 2.

As depicted in Fig. 3, the coupled fire risk model A-Star algorithm considers the effect of fire and its products on the safety of the path to a certain extent in path planning. The planned path effectively avoids the area affected by the fire and maintains a certain distance from the fire source to ensure the safety of the path, though it is not the shortest evacuation path in geometric length.

Table 2

Pseudocode of coupled fire risk A-Star algorithm.

Algorithm 2: Coupled fire risk A-Star algorithm ($P[start] \rightarrow P[dest]$)	
1:	Mark $P[start]$ as <i>openlist</i> .
2:	while <i>openlist</i> $\neq$ empty do
3:	Select the node $P[i]$ from the <i>openlist</i> whose value of evaluation function $F(P[i])$ is smallest.
4:	if $P[i] = P[dest]$ then
5:	return "path is found".
6:	else
7:	Mark $P[i]$ as <i>closelist</i> .
8:	if Successor $P_i[j]$ of node $P[i]$ is not belong to <i>closelist</i> or <i>obstacle</i> node then
9:	Mark $P_i[j]$ as <i>openlist</i>
10:	Calculate $\xi(j)$. Then, calculate the equivalent length $L(j)$ of nodes $P[i]$ to $P_i[j]$.
11:	if $P_i[j]$ belongs to <i>openlist</i> and $F(P_i[j])$ is smaller than $F(P_m[j])$ when $P[m]$ was marked as <i>closelist</i> then
12:	Set parent node of $P[j]$ as $P[i]$, and $F(P[i]) = F(P_i[j])$.
13:	end if
14:	end if
15:	end if
16:	end while
17:	return "the path cannot be found".

**Fig. 3.** Coupled fire risk A-Star algorithm path planning results.

3.3. Dynamic avoid-smoke A-Star algorithm

The traditional A-Star algorithm is a path planning algorithm for static environment information, where the search environment information does not change with time. In the metro station fire scene, the fire and its influence are constantly changing with time. It is impossible to meet the safety requirements of evacuation only based on the static environmental information at a certain time. Accordingly, the algorithm for evacuation path planning in fire must have the ability to deal with dynamic fire environment information. Based on this, a dynamic update mechanism is introduced and based on Section 3.2, the Dynamic Avoid-Smoke A-Star algorithm is proposed for dynamic planning of evacuation paths in metro station fire scenarios.

3.3.1. Dynamic update mechanism

The traditional A-Star algorithm can only solve the static obstacle problem. When the current node $P[i]$ selects the next node in path planning, the state information of the surrounding eight child nodes should be first determined. Next, the state information of the surrounding child nodes is fixed and does not change with time. In the metro station fire scenario, the flame spread will lead to the formation of

novel combustion areas in the metro station, and the above-mentioned areas dynamically vary with time. For evacuation path planning, the novel combustion area acts as a dynamic obstacle to the passage of people. Thus, the algorithm for evacuation path planning in metro stations should be capable of coping with dynamic obstacles. To address dynamic obstacles, the planned path should exhibit temporal properties to achieve avoidance of dynamic obstacles in time and space dimensions, whereas the trajectory of the path planned by the algorithm is not required to intersect with the changing trajectory of dynamic obstacles.

This study conforms to the optimization of the traditional A-star algorithm. The A-star algorithm follows the current section $P[i]$ for searching. The core of the dynamic update mechanism refers to determining the time T_i from the starting point $P[start]$ to the current node $P[i]$. The subprogram is termed 'Dynamic-update', and its pseudocode is listed in Table 3:

The algorithm takes different times when choosing to move along a straight line or a diagonal line in the raster map, because more path costs are consumed when moving along a diagonal line. Accordingly, the time T_i of the current node $P[i]$ in the raster map cannot be determined based on the number of grids, and the time of the current node should be determined based on the distance from the starting point to the current node divided by the speed.

The subroutine Dynamic-update calculates the time T_i of the current node $P[i]$ by superimposing the time of each move. It is worth noting that the time T_n of a node, when its parent node changes, the time of that node also changes and there are cases where the nodes are different but have the same time. Since the spreading of flames will form new obstacles, the path planning is conducted using the A-star algorithm based on a dynamic update mechanism. In path planning, when the current node $P[i]$ performs node selection, whether the child nodes of the current node $P[i]$ are the new burning area at the current moment T_i will be judged. If so, it will be considered the obstacle node $D_n[T_i]$ and then introduced to the *closelist*, such that the rest of the children nodes of the current node $P[i]$ are substituted into the *openlist*, its $F(P[i])$ value is calculated, and the node with the smallest F value is selected from the *openlist*. The relevant path planning process is shown in Fig. 4. As shown in Fig. 4(a), the flame burning range is small, node (2, 3) serves as the current node, and the current time is T_1^* . The F value of the current node's child nodes is calculated and added to the *openlist*. The fire source location (2, 4) node is introduced as an obstacle to the *closelist* for the next node selection. On that basis, the node (3, 3) in the *openlist* as the point with the smallest F value serves as the current node, the current time is T_2^* , the flame burning range is expanded at this moment involving the nodes (1, 4), (2, 4), (3, 4) (Fig. 4(b)). The above result is considered an obstacle node and added to the *closelist*. The F value of the

Table 3

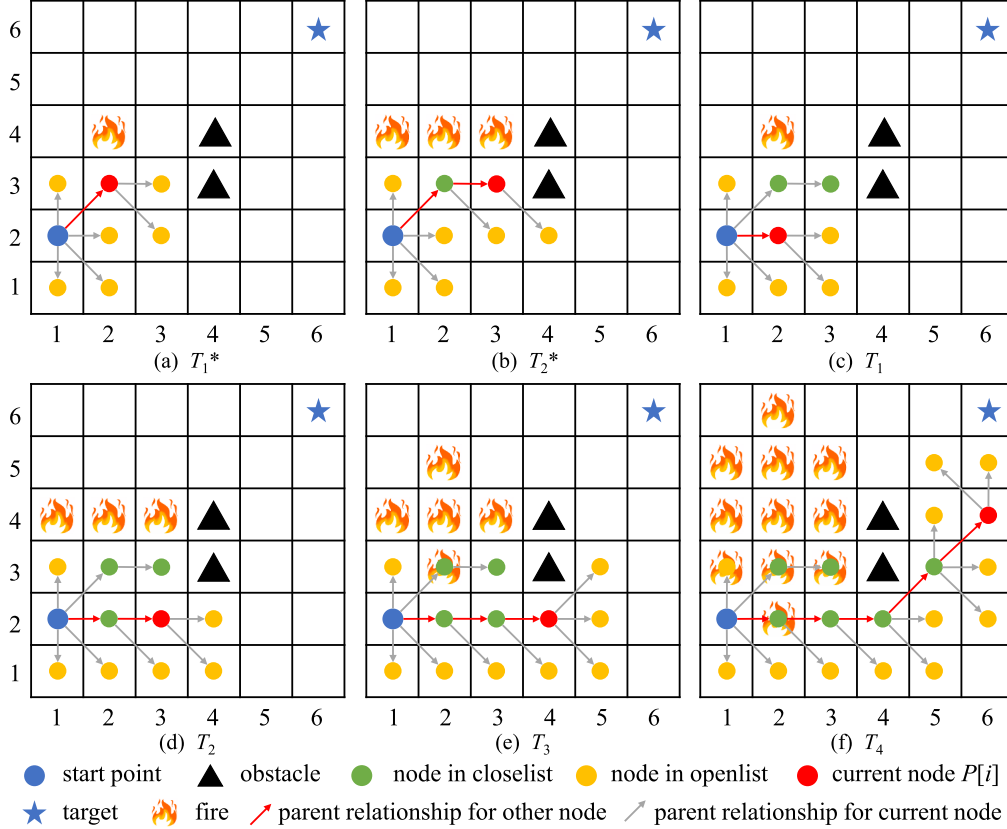
Pseudocode of Dynamic-update.

Algorithm 3: Dynamic-update ($P[start]$, $P[i]$)

```

1:  $T_i = 0$ 
2: while  $P[i] \neq P[start]$  do
3:   Find the parent node  $P[i - 1]$  of the  $P[i]$ .
4:    $T_i = T_i + \text{cost time from } P[i - 1] \text{ to } P[i]$ 
5:    $P[i] = P[i - 1]$ 
6: end while
7: return  $T_i$ 

```

**Fig. 4.** The search process of A-star algorithm with dynamic update mechanism.

current node's child is determined and then added to *openlist* for the next node selection. As the node (3, 4) at this moment has been involved in fire, it cannot serve as a path node, such that the node with the smallest F in the *openlist* is (2, 2). The parent node of node (2, 2) represents the starting point (1, 2) and the time update of the current node (2, 2) is expressed as T_1 (Fig. 4(c)), the F of the children of the current node is determined and added to the open list for the next node selection. On that basis, the node (3, 2) in the *openlist* acts as the current node as the point with the smallest F value and the current time is T_2 (Fig. 4(d)), the F value of the children of the current node is determined and then introduced to the *openlist* for the next node selection. Likewise, as shown in Fig. 4(e). In Fig. 4(f), with the development of the fire, at the moment of T_4 fire, although the node (2, 2) is involved, the current node (6, 5) is not involved, such that the next node can be selected.

The pseudocode of the A-star algorithm with dynamic update mechanism is shown in Table 4, where the algorithm determines the final path planning result by backtracking.

3.3.2. Principle of Dynamic Avoid-Smoke A-Star algorithm

This section establishes a dynamic planning algorithm for evacuation paths in an underground station fire scenario, based on the coupled fire

risk A-Star algorithm built in section 3.2 combined with a dynamic update mechanism, the principles of the algorithm are as follows.

The temperature, visibility, and CO concentration in the fire environment change over time, so the path equivalent length under the effect of fire will also change with time, which makes the fire risk cost coefficient have time attribute. The fire risk cost coefficient (Eq. (4)) is updated as:

$$\xi_t = \frac{L_t}{l} = \frac{1}{f_t(T, K_c, \rho_{co})} \quad (13)$$

where ξ_t represents the fire risk cost coefficient at time t , L_t represents the equivalent length of the evacuation path at time t , $f_t(T, K_c, \rho_{co})$ represents the fire smoke effect function at time t .

The practical cost function in the evaluation function of the DASA-Star algorithm to satisfy the consideration of the fire risk of the path over time is defined as:

$$G_t(P[i]) = \sum_{n=1}^k \sum_{t_0}^{t_i} l(n) \xi_t(n) = \sum_{n=1}^k \sum_{t_0}^{t_i} L_t(n) \quad (14)$$

where t_i denotes the time of the current node $P[i]$, t_0 represents the time

Table 4

Pseudocode of A-star algorithm with dynamic update mechanism.

Algorithm 4: Dynamic A-star algorithm ($P[start] \rightarrow P[dest]$)	
1:	Mark $P[start]$ as <i>openlist</i> .
2:	while <i>openlist</i> $\neq$ empty do
3:	Select the node $P[i]$ from the <i>openlist</i> whose value of evaluation function $F(P[i])$ is smallest.
4:	if $P[i] = P[dest]$ then
5:	return “path is found”.
6:	else
7:	Mark $P[i]$ as <i>closelist</i> .
8:	$T_i = \text{Dynamic-update}(P[start], P[i])$
9:	if $P_i[j]$ of node $P[i]$ is fire at time T_i .
10:	Mark $P_i[j]$ as dynamic obstacle $D_n[T_i]$, and $P_i[j]$ as <i>closelist</i> .
11:	end if
12:	if Successor $P_i[j]$ of node $P[i]$ does not belong to <i>closelist</i> or <i>obstacle</i> node then
13:	Mark $P_i[j]$ as <i>openlist</i> and calculate $F(P_i[j])$.
14:	if $P_i[j]$ belongs to <i>openlist</i> and $F(P_i[j])$ is smaller than $F(P_n[j])$ when $P[n]$ was marked as <i>closelist</i> then
15:	Set parent node of $P[j]$ as $P[i]$, and $F(P[i]) = F(P_i[j])$.
16:	end if
17:	if $D_n[T_i]$ does not belong to <i>openlist</i> then
18:	Remove $D_n[T_i]$ from <i>closelist</i> .
19:	end if
20:	end if
21:	end if
22:	end while
23:	return “the path cannot be found”.

of $P[start]$ node, that is, the time when the algorithm starts searching.

Then the evaluation function $F_t(P[i])$ of the DASA-Star algorithm is defined as:

$$F_t(P[i]) = G_t(P[i]) + H(P[i]) \quad (15)$$

The pseudocode of the DASA-Star algorithm proposed in this section is shown in Table 5.

4. Case study

In the present section, simulation experiments are performed to verify the effectiveness of the proposed DASA-Star algorithm. Intel i5 series 1.8 GHz processor with 8 GB memory is employed as the simulation platform, and MATLAB R2016b under Windows 10 and FDS 6.7.1 serves as the simulation software. First, the metro station model is built using the fire simulation software FDS. Besides, the fire simulation of the

metro station hall is conducted to acquire the relevant fire smoke data. Second, the traditional A-Star algorithm, the coupled fire risk A-Star algorithm, and the DASA-Star algorithm are employed for evacuation path planning in the metro station fire scenario, respectively, based on the simulation results of FDS. Lastly, the differences between the evacuation path planning results of a variety of algorithms are compared, and the path planning results of different algorithms are evaluated to verify the effectiveness of the algorithms.

4.1. Metro station fire simulation

Fig. 5 presents the layout of the building structure at the lobby level of the metro station, taking up a total area of 3039.75 m² in the north-south direction, which comprises four safety exits, with a floor height of 3 m, a grid size of 0.5 × 0.5 × 0.2 m, as well as a total grid number of 182,385. The fire source (1 m × 1 m) is set at one end of the

Table 5

Pseudocode of DASA-Star algorithm.

Algorithm 5: Dynamic Avoid-Smoke A-Star algorithm ($P[start] \rightarrow P[dest]$)	
1:	Mark $P[start]$ as <i>openlist</i> .
2:	while <i>openlist</i> $\neq$ empty do
3:	Select the node $P[i]$ from the <i>openlist</i> whose value of evaluation function $F_t(P[i])$ is smallest.
4:	if $P[i] = P[dest]$ then
5:	return “path is found”.
6:	else
7:	Mark $P[i]$ as <i>closelist</i> .
8:	$T_i = \text{Dynamic-update}(P[start], P[i])$
9:	if $P_i[j]$ of $P[i]$ satisfies any of the following conditions at time T_i : $T > 120^\circ\text{C}$ or $K_c > 1.2$ or $\rho_{co} > 0.1\%$.
10:	Mark $P_i[j]$ as dynamic obstacle $D_n[T_i]$, and $P_i[j]$ as <i>closelist</i> .
11:	end if
12:	if Successor $P_i[j]$ of node $P[i]$ does not belong to <i>closelist</i> or <i>obstacle</i> node then
13:	Mark $P_i[j]$ as <i>openlist</i> .
14:	Calculate the ξ_i value of $P_i[j]$ at time T_i by Eq. (13), to calculate the equivalent length $L_e(j)$ of nodes $P[i]$ to $P_i[j]$. Then, calculate $F_t(P_i[j])$.
15:	if $P_i[j]$ belongs to <i>openlist</i> and $F_t(P_i[j])$ is smaller than $F_t(P_n[j])$ when $P[n]$ was marked as <i>closelist</i> then
16:	Set parent node of $P[j]$ as $P[i]$, and $F_t(P[i]) = F_t(P_i[j])$.
17:	end if
18:	if $D_n[T_i]$ does not belong to <i>openlist</i> then
19:	Remove $D_n[T_i]$ from <i>closelist</i> .
20:	end if
21:	end if
22:	end if
23:	end while
24:	return “the path cannot be found”.

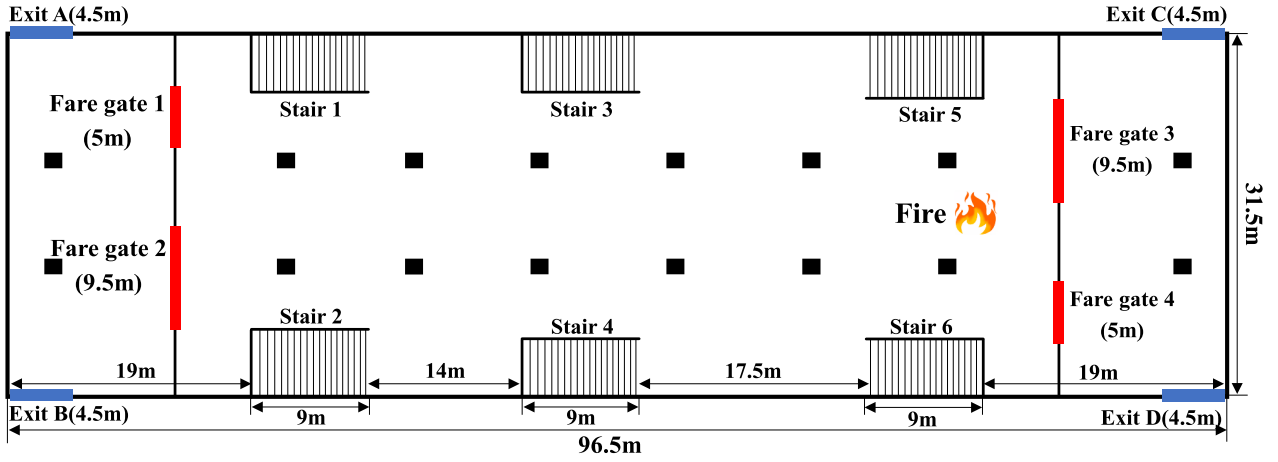


Fig. 5. Layout of metro station hall floor.

paid area of the metro station, and the burning item is set as the luggage brought in by passengers, with polyurethane as the main component. The fire source heat release rate HRR is set as a fast type, and the growth coefficient α is taken as 0.04689, the fire heat release rate is set as 6 MW, and the fire heat release rate equation shows that the fire source reaches the maximum heat release rate at the 357 s (Kadokura et al., 2012; Yan et al., 2023). The average height of the human eye is 1.6 m (Xu et al., 2022). To ensure the safety of occupant evacuation, this study monitored the numerical changes of temperature, CO concentration and visibility at 1.8 m.

4.2. Algorithm simulation experiment

The metro station has a long and narrow spatial layout. When the fire occurs at one end of the station, the fire smoke slightly affects the end far from the fire source, such that the local scene of the end with the fire source in the metro station is selected in this study for path planning.

Notably, the A-Star algorithm is usually adopted to raster map path solving, where 3D objects are downscaled to 2D planes based on environmental information, and a raster is employed to compose the search map. The selected subway station scene is rasterized. Selecting a larger grid size will increase the risk of generating a path, and selecting a smaller grid size will lead to a long algorithm planning time. In order to

ensure the accuracy of path planning and the real-time performance of algorithm planning, it is also necessary to meet the minimum requirements of a single person. The size of the grid unit is $0.5\text{ m} \times 0.5\text{ m}$, and the grid size is 76×63 . The end point of the evacuation path planning is set at a point $P[\text{dest}] = (72, 60)$ close to safety exit C, and the starting point is set at a point $P[\text{start}] = (11, 7)$ near escalator #6 at the station concourse level.

4.2.1. Analysis of path planning results of coupled fire risk A-Star algorithm

Given the limited effect range of fire at the early stage of fire development, the fire environment information based on 50 s time is selected to clearly show the effect of fire on the evacuation environment, and the traditional A-Star algorithm, the coupled fire risk A-Star algorithm and the risk-free A-Star algorithm are adopted to plan the path respectively. The path planning results are shown in Fig. 6, and the geometric length of the path and the sum of fire risks are compared as shown in Fig. 7.

In the process of path planning, the traditional A-Star algorithm can quickly calculate the shortest path with the shortest geometric length of 41.471 m based on the principle of minimizing the geometric length of the path. Since the traditional A-Star algorithm does not consider the effect of fire on the safety of the path in path planning and only pursues the shortest geometric length of the path, which leads to a higher fire

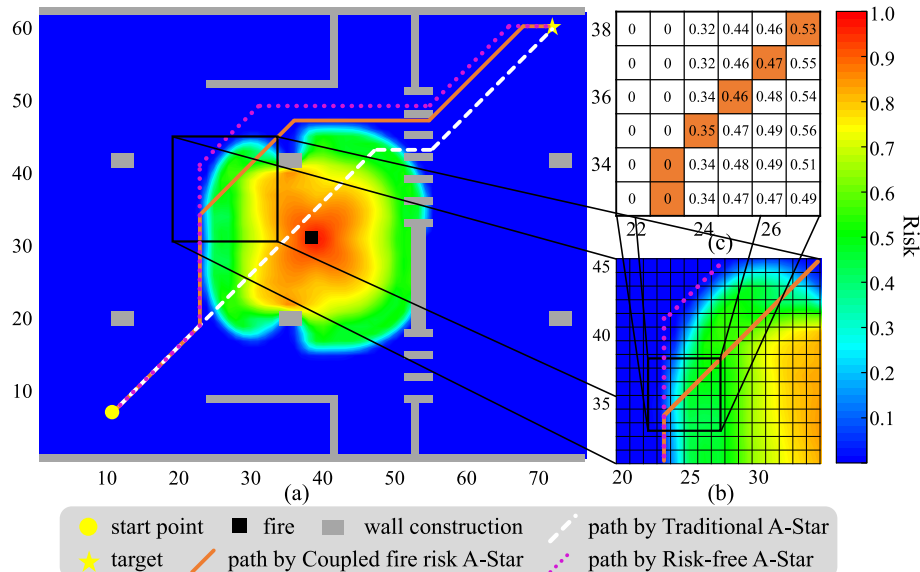


Fig. 6. Comparison of the path planning results of the traditional A-Star algorithm, risk-free A-Star algorithm and coupled fire risk A-Star algorithm.

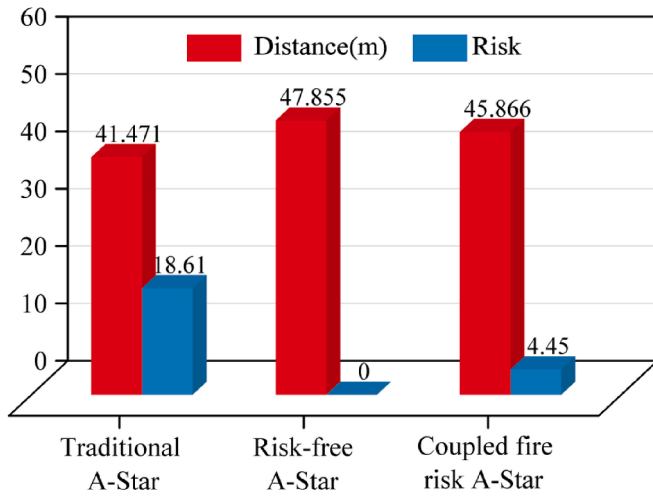


Fig. 7. Comparison of geometric lengths and fire risks of paths of the traditional A-Star algorithm, the risk-free A-Star algorithm, and the coupled fire risk A-Star algorithm.

risk in the planned path part stage close to the fire source, and the fire smoke risk value in the path part stage exceeds 0.8, as shown in Fig. 6(a). The total fire risk of the path is 18.61, as shown in Fig. 7. Compared with the traditional A-Star algorithm, the Risk-free A-star algorithm considers the effect of fire on the path safety maximally, and it does not involve areas where fire exerts a negative effect. The fire risk of the entire path is 0, as shown in Fig. 6(a). Although the algorithm avoids the area where the fire risk is not 0, it costs more for the path, and the path has a geometric length of 47.855 m, as shown in Fig. 7. The geometric length of the path planned by the coupled fire risk A-Star algorithm is 45.866 m, and the total fire risk of the path is 4.45, as shown in Fig. 7. Although the path planned by the algorithm does not completely avoid the area where the fire exerts a negative effect, and some stages involve areas where the fire has a negative impact. However, compared with the path planned by the traditional A-Star algorithm, the total fire risk of the path is reduction 76 %, as shown in Fig. 7.

4.2.2. Analysis of algorithm path planning results under fire information at different times

Set the same start and end points. The traditional A-Star algorithm, the coupled fire risk A-Star and the risk-free A-Star algorithm are employed for path planning based on the fire scene information at 35 s, 45 s, and 55 s. Fig. 8 presents the local path of the path planning results of the three traditional A-Star algorithms at different times. Fig. 9(a, b) presents the statistics of the geometric length of the path planned by the

algorithm in accordance with the fire information at different times and the sum of the fire risk of the path.

As shown in Fig. 8, the path planning results of the traditional A-Star algorithm do not change under the fire environment information at different times. The traditional A-Star algorithm will not change the path planning result under the change of fire environment information. The path planning result is fixed when the starting point, the end point, and obstacles are fixed. However, the coupled fire risk A-Star algorithm and the risk-free A-Star algorithm consider the effect of fire on evacuation, and their path planning results will change in accordance with the change of fire environment information. The coupled fire risk A-Star algorithm aims at finding the path with the shortest equivalent length under the effect of fire. The equivalent length of the path under fire varies with fire environment information, such that the path planning results of the algorithm are also different, as shown in Fig. 8. The Risk-free A-Star algorithm is capable of planning the shortest path without fire risk to avoid the fire-affected area as shown in Fig. 8(a, b). However, when the effect range of fire expands to a certain extent, no path will be likely to be from the start to the end, and the risk-free A-Star algorithm does not yield any solutions, as shown in Fig. 8(c).

The geometric length of the path planned by the traditional A-Star algorithm is the shortest compared with the coupled fire risk A-Star algorithm and the Risk-free A-Star algorithm. However, with the development of fire, the fire risk of the path with fixed path planning results will be increased continuously, as shown in Fig. 9(a). The coupled fire risk A-Star algorithm will sacrifice the geometric length of the path to avoid the area with higher fire risk. When the effect range and effect degree of the fire increase with time, the geometric length of the path will be increased. However, the total fire risk of the planned path maintained in a low range, over 65 % lower than that of the traditional A-Star algorithm, as shown in Fig. 9(b). Although the risk-free A-Star algorithm plans a path with a full fire risk of 0, the geometric length of the path is the longest compared with the traditional A-Star algorithm and the coupled fire risk A-Star algorithm. In brief, the coupled fire risk A-Star algorithm conforms to the fire environment information at a certain time, and the planned path is capable of effectively avoiding the area with high fire risk, which is the shortest path of equivalent length under the effect of fire at that time.

4.2.3. Analysis of path planning results of DASA-Star algorithm

Evacuation path planning is conducted in combination with fire simulation results to evaluate the effectiveness of the DASA-Star algorithm for dynamic planning of evacuation paths in fire scenarios. Nearly 150 s after the fire, the temperature of the fire source adopted to escape from the gate units 3 and 4 approaches 120°C. Since the evacuation end point is set close to the safety exit C, it should pass through the gate units 3 and 4 to reach the end point. To ensure that the occupant is capable of

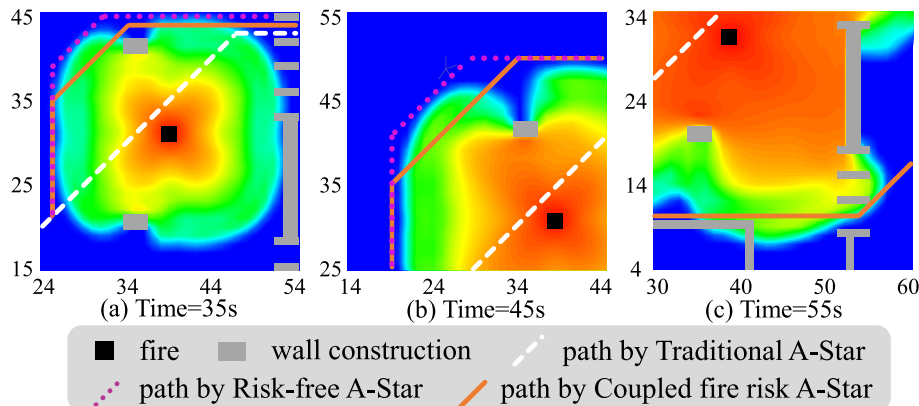


Fig. 8. Comparison of the path planning results of the traditional A-Star algorithm, the risk-free A-Star algorithm, and the coupled fire risk A-Star algorithm at different times.

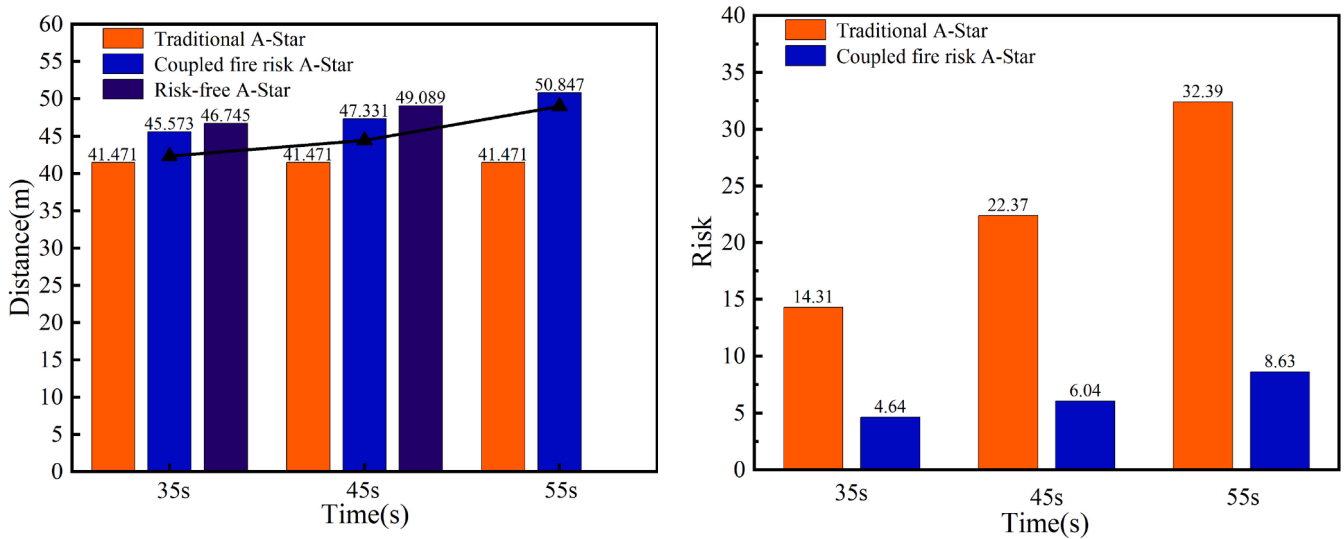


Fig. 9. The geometric length and fire risk of the path planned by different algorithms at different times.

passing through the gate units 3 and 4 safely to reach the end point (DASA-Star algorithm path planning has a solution), the evacuation path planning is conducted at the early stage of the fire, slightly threatening the gate units 3 and 4. The DASA-Star algorithm and the dynamic A-Star algorithm are adopted to plan the evacuation path at 20 s after the fire. The idealized moving speed of personnel in the algorithm planning process reaches 1.2 m/s (Li et al., 2020b; Zhang et al., 2022b). In addition, based on the fire scene information at the start time of 20 s, the path planning is conducted using the coupled fire risk A-Star algorithm, and the path planning results are shown in Fig. 10.

Fig. 11 illustrates the path planning process of the DASA-Star algorithm and the dynamic A-Star algorithm. As shown in Fig. 11(a), the DASA-Star algorithm has planned a part of the path at 28.7 s. At this moment, the DASA-Star algorithm abandons the path with the shortest geometric length while bypassing the area where the fire adversely affects evacuation. The above-mentioned decision leads to the significantly increased geometric length of the path, whereas the subsequent planned path avoids the area with high fire risk, which can be ensured. When the calculation reaches 34.2 s, the DASA-Star algorithm changes its direction while beginning to enter the area where the fire adversely

affects evacuation, whereas it will select nodes with fire risk in a safe range to approach the end point, as shown in Fig. 11(b). At 43.5 s, due to the limitation of the position of the gate unit, the algorithm changes the path planning direction again. It is expected to select the farthest side from the fire source point to pass through the gate unit 3 and tend to the end point, as shown in Fig. 11(c). In addition, at 49.5 s, the current node of the DASA-Star algorithm tends to the end point in the same direction of fire smoke diffusion, and it begins to stay away from the area where the fire adversely affects the evacuation, as shown in Fig. 11(d). Thus, when the current node performs path selection, the shortest geometric length of the path is the main principle. The path planned by the dynamic A-Star algorithm reaches the end point safely at 54.6 s, and the path planned by the DASA-Star algorithm reaches the end point safely at 57.7 s, as shown in Fig. 11(e, f).

Fig. 12(a, b) shows the calculation time of each algorithm (30 test cycles), the length of the planned path and the sum of the fire risk of the path. Fig. 13 compares the fire risks involved in path planning of the dynamic A-Star algorithm and the DASA-Star algorithm. The dynamic A-Star algorithm still takes the shortest geometric length of the path as the only decision principle, and the planned path has the shortest geometric length. However, the algorithm does not consider the effect of fire in path planning, such that the total fire risk of the path in the planning process reaches up to 20.09. The geometric length of the path planned by the DASA-Star algorithm is 45.28 m, and the total fire risk of the path reaches 6.67. Compared with the path planning results of the dynamic A-Star algorithm, the geometric length of the path is increased by 9.2 %, the total fire risk of the path declines by 66.8 %. The average fire risk of nodes in the area where fire adversely affects evacuation in path planning of the dynamic A-Star algorithm and the DASA-Star algorithm is 0.72 and 0.51, respectively. In the path process of the DASA-Star algorithm, although some stage paths fall into the area where fire adversely affects evacuation, the number of nodes involved is lower than that of the dynamic A-Star algorithm, and the average fire risk of the nodes involved declines by 29.5 %. The path planned by the A-Star algorithm based on the fire environment information at 20 s achieves a geometric length of 42.643 m, shorter than the geometric length of the path planned by the DASA-Star algorithm. However, at 57.7 s, the total fire risk of the path reaches 40.37, significantly higher than the path planned by the DASA-Star algorithm. In brief, the DASA-Star algorithm considers the effect of the dynamic development of the fire on the evacuation process, and the planned path is capable of avoiding the high fire risk in the evacuation process effectively. However, compared with other algorithms, the DASA-Star algorithm adds a dynamic update mechanism

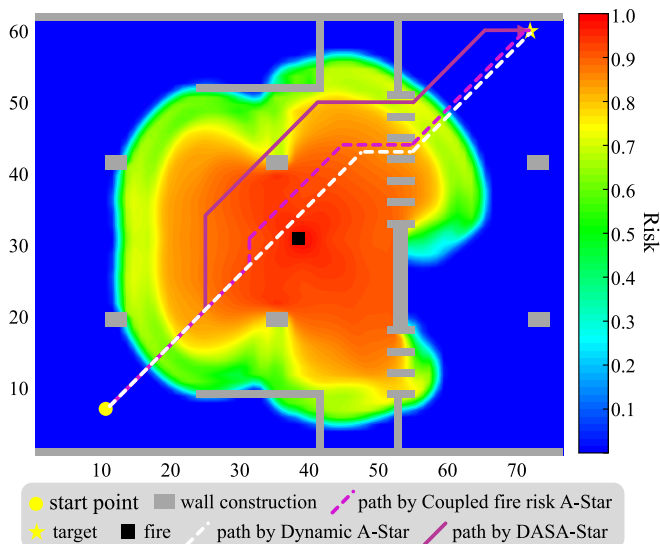


Fig. 10. Comparison of the dynamic A-Star algorithm, the DASA-Star algorithm, and 20 s coupled fire risk A-Star algorithm.

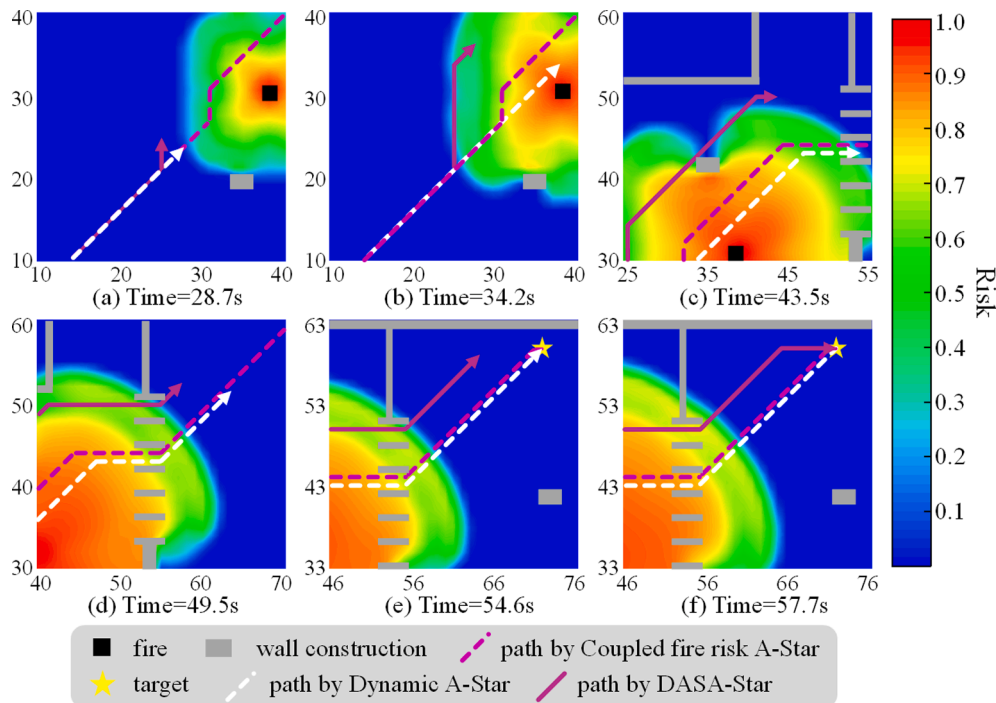


Fig. 11. Path planning of dynamic A-Star algorithm and DASA-Star algorithm.

and a fire risk model to ensure the safety of the planned path while greatly increasing the complexity and calculation of the search process. The search time of the DASA-Star algorithm is higher than that of other algorithms.

5. Discussions

Without considering the effect of fire development and change on evacuation during evacuation, through the simulation results of traditional A-Star algorithm, coupled fire risk A-Star algorithm and risk-free A-Star algorithm, it can be seen that compared with the path planned by traditional A-Star algorithm, the geometric length of the path planned by coupled fire risk A-Star algorithm increases, but the total fire risk of the path can be reduced by more than 65 %, which makes the path safer in fire conditions. Compared with the risk-free A-Star algorithm, the geometric length of the path is shorter. Although the fire risk of the path is not all 0, the path involves the stage of the negative impact area of the fire, and the fire risk value is below 0.6. It can be seen that based on the fire environment information at a certain time, the coupling fire risk A-Star algorithm can plan the optimal path. However, the coupled fire risk A-Star algorithm only considers static fire information, whereas the development and change of fire in the evacuation process are ignored. The planned path is recognized as the optimal path at this time, whereas the practical evacuation refers to a process with time attribute. Accordingly, the path planned by the coupled fire risk A-Star algorithm cannot conform to the practical evacuation needs.

However, the DASA-Star algorithm considers the effect of the dynamic development of the fire on the evacuation process. In path planning, the moving speed of the occupants is introduced, and the environmental information is updated in real time in accordance with the time of the current node, with the aim of determining the real-time optimal path. Compared with the path planned by the dynamic A-Star algorithm, the average fire risk of the path planned by the DASA-Star algorithm declines by 29.5 %, and the total fire risk of the path is lowered by 66.8 %. The above-described method can effectively avoid the area with high fire risk in the evacuation process; it is also the path with the shortest equivalent length under the effect of fire. The DASA-Star algorithm is capable of planning safer and more effective paths in

evolving fire scenarios to conform to practical evacuation requirements.

6. Conclusion

Based on the A-Star algorithm, this study proposes the DASA-Star algorithm to solve the problem that the traditional A-Star algorithm cannot realize the dynamic planning of the evacuation path and cannot consider the negative effect of the fire. The search principle of the traditional A-Star algorithm and its limitations in path planning in fire scenarios are analyzed. The effect of temperature, visibility and CO concentration in the environment on the evacuation speed is further analyzed, and the fire risk model is built to calculate the equivalent length of the path under the effect of fire. The dynamic update mechanism considering time factor is introduced, which makes the algorithm to cope with the dynamic changes of environmental information. Combined with the results of FDS fire simulation, the results of algorithm simulation experiments suggest that the DASA-Star algorithm proposed in this study consider the negative effect of fire on evacuation, and the planned path effectively avoids the dangerous area with high fire risk in the fire scene. Compared with the traditional A-Star algorithm, the total fire risk of the path can be reduced by over 65 %, such that the safety of the path can be significantly increased, and the algorithm considers the development and change of fire during the movement of personnel, and introduces a dynamic update mechanism. In path planning, the evacuation environment information is updated in real time according to the fire simulation results, so as to be close to the practical evacuation process and make the planned path is more effective.

7. Limitation discussions

Several limitations remain in this study. When establishing an evacuation path planning model, in addition to considering the impact of the fire itself on evacuation, it is also necessary to consider the impact of personnel interaction and psychological fluctuations on evacuation during the evacuation process. When setting the evacuation reference speed, differences should also be made according to the age and gender of the individual population. In addition, the premise of the evacuation

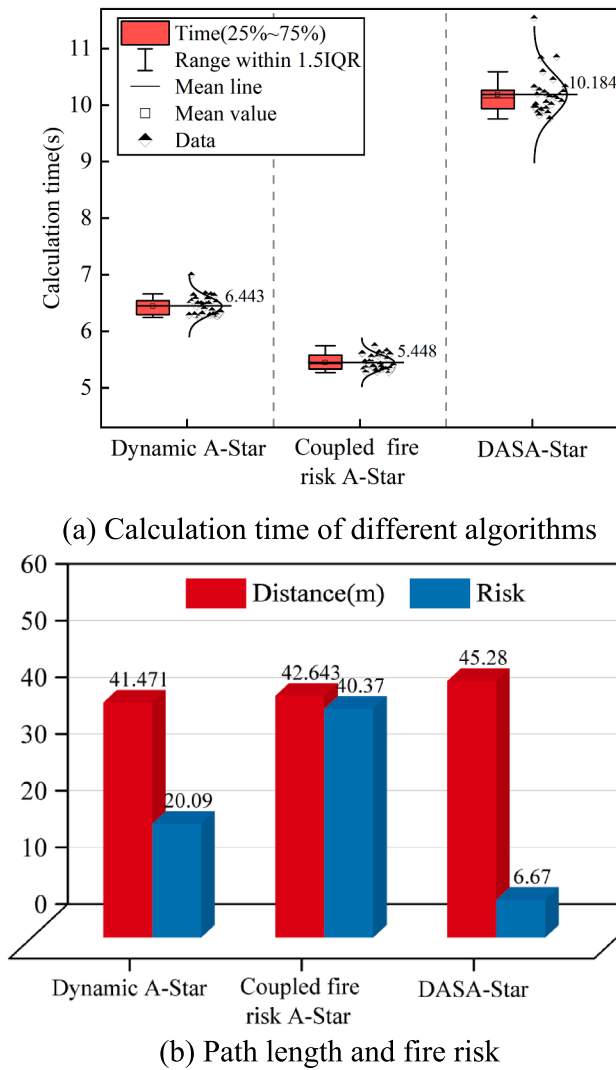


Fig. 12. Comparison of calculation time, path length and fire risk of different algorithms.

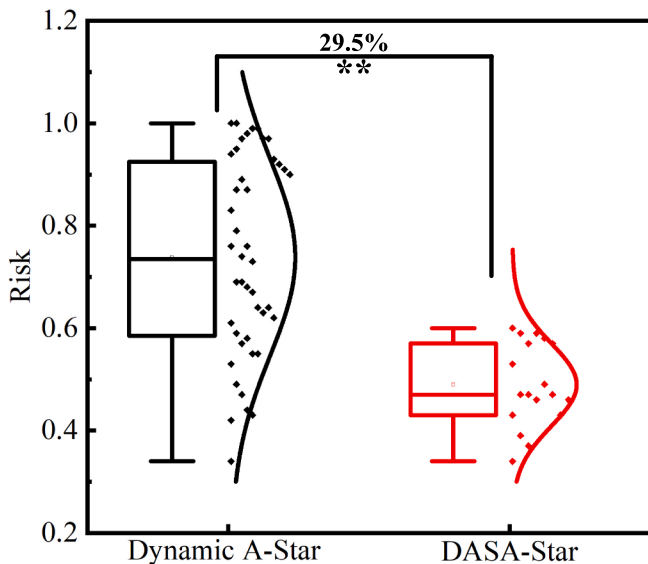


Fig. 13. The fire risk involved in path planning of Dynamic A-Star algorithm and DASA-Star algorithm.

path planning model established in this study is that the development process of the fire is known. However, there are many uncertainties in the real fire, and the existing fire simulation software still cannot predict the development process of the real fire in advance.

CRediT authorship contribution statement

Songtao Zuo: Writing – original draft, Validation, Software, Methodology, Formal analysis. **Zhanli Mao:** Writing – review & editing, Software, Methodology, Formal analysis, Conceptualization. **Chuangang Fan:** Writing – review & editing. **Xin Chen:** Supervision, Formal analysis. **Meiling Gong:** Validation, Formal analysis. **Jianxing Ren:** Software, Formal analysis. **Xiaoxiao Fan:** Formal analysis. **Yongnan Guo:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Bagheri, S.M., Taghaddos, H., Mousaei, A., Shahnavaz, F., Hermann, U., 2021. An A-Star algorithm for semi-optimization of crane location and configuration in modular construction. *Autom. Constr.* 121.
- Cai, Z., Zhou, R., Cui, Y., Wang, Y., Jiang, J., 2022. Influencing factors for exit selection in subway station evacuation. *Tunn. Undergr. Space Technol.* 125.
- Chen, S., Di, Y., Li, F., Xu, Q., Liu, H., 2017. Passenger evacuation model of metro platform considering psychological stress. *J. Traffic Transp. Eng.* 17, 113–120.
- Choi, M., Chi, S., 2019. Optimal route selection model for fire evacuations based on hazard prediction data. *Simul. Model. Pract. Theory* 94, 321–333.
- Coskun, G., Demir, U., Soyhan, H.S., 2022. Investigation of the smoke ventilation and evacuation strategies to decrease smoke poisoning risk by coupling fire and evacuation simulations. *J. Appl. Fluid Mech.* 15.
- Cuesta, A., Abreu, O., Balboa, A., Alvear, D., 2017. Real-time evacuation route selection methodology for complex buildings. *Fire Saf. J.* 91, 947–954.
- Fang, Z., Song, W., Zhang, J., Wu, H., 2012. A multi-grid model for evacuation coupling with the effects of fire products. *Fire Technol.* 48, 91–104.
- Fridolf, K., Ronchi, E., Nilsson, D., Frantzi, H., 2019. The representation of evacuation movement in smoke-filled underground transportation systems. *Tunn. Undergr. Space Technol.* 90, 28–41.
- Galea, E.R., Xie, H., Deere, S., Cooney, D., Filippidis, L., 2017. An international survey and full-scale evacuation trial demonstrating the effectiveness of the active dynamic signage system concept. *Fire Mater.* 41, 493–513.
- Gibson, J., Schuler, T., McGuire, L., Lofaro, D.M., Sofge, D., 2020. Swarm and multi-agent time-based A* path planning for lighter-than-air systems. *Unmanned Syst.* 08, 253–260.
- Han, Y., Liu, H., Moore, P., 2017. Extended route choice model based on available evacuation route set and its application in crowd evacuation simulation. *Simul. Model. Pract. Theory* 75, 1–16.
- Hart, P.E., Nilsson, N.J., Raphael, B., 1968. A formal basis for the heuristic determination of minimum cost paths. *IEEE Trans. Syst. Man Cybern. Syst.* 4, 100–107.
- He, Z., Shen, K., Lan, M., Weng, W., 2023. An evacuation path planning method for multi-hazard accidents in chemical industries based on risk perception. *Reliab. Eng. Syst. Saf.* 244.
- He, Z., Liu, C., Chu, X., Negenborn, R.R., Wu, Q., 2022. Dynamic anti-collision A-star algorithm for multi-ship encounter situations. *Appl. Ocean Res.* 118, 102995.
- Helbing, D., Farkas, I., Vicsek, T., 2000. Simulating dynamical features of escape panic. *Nature* 407, 487–490.
- Huang, P., Chen, M., Chen, K., Ye, S., Yu, L., 2022. Study on an emergency evacuation model considering information transfer and rerouting: taking a simplified H-shape metro station hall as an example. *Tunn. Undergr. Space Technol.* 124.

- Huang, X., Jin, J., Lin, Z., Che, L., Liu, J., 2021. Dynamic evacuation path planning for fire disaster of deep underground space based on A* algorithm. *J. B Univ. Technol.* 47, 702–709.
- Huang, D., Lo, S.M., Chen, J., Fu, Z., Zheng, Y., Luo, L., Zhuang, Y., Cheng, H., Yang, L., 2019. Mapping fire risk of passenger-carried fire load in metro system via floor field cellular automaton. *Autom. Constr.* 100, 61–72.
- Kadokura, H., Sekizawa, A., Takahashi, W., 2012. Study on availability and issues of evacuation using stopped escalators in a subway station. *Fire Mater.* 36, 416–428.
- Khakzad, N., 2023. A methodology based on Dijkstra's algorithm and mathematical programming for optimal evacuation in process plants in the event of major tank fires. *Reliab. Eng. Syst. Saf.* 236.
- Kim, M., Ham, Y., Koo, C., Kim, T.W., 2023. Simulating travel paths of construction site workers via deep reinforcement learning considering their spatial cognition and wayfinding behavior. *Autom. Constr.* 147.
- Li, T., Chen, J., Ma, J., Lo, S.M., 2016. Experimental study of passenger evacuation efficiency in a metro station. *J. Transp. Syst. Eng. Inf. Technol.* 16, 228–234.
- Li, Y., Chen, M., Zheng, X., Dou, Z., Cheng, Y., 2020a. Relationship between behavior aggressiveness and pedestrian dynamics using behavior-based cellular automata model. *Appl. Math. Comput.* 371.
- Li, Z., Huang, H., Li, N., Chu, M.L., Law, K., 2020b. An agent-based simulator for indoor crowd evacuation considering fire impacts. *Autom. Constr.* 120.
- Liang, C., Zhang, X., Watanabe, Y., Deng, Y., 2021. Autonomous collision avoidance of unmanned surface vehicles based on improved A Star and minimum course alteration algorithms. *Appl. Ocean Res.* 113, 102755.
- Liu, M., Zhang, F., Ma, Y., Pota, H.R., Shen, W., 2016. Evacuation path optimization based on quantum ant colony algorithm. *Adv. Eng. Inform.* 30, 259–267.
- Lo, S.M., Fang, Z., 2000. A spatial-grid evacuation model for buildings. *J. Fire Sci.* 18, 376–394.
- Lo, S., Ming, 1996. An evacuation model for buildings: a systems approach, HKU Theses Online (HKUTO). The University of Hong Kong (Pokfulam, Hong Kong).
- Milke, J.A., 2000. Evaluating the early development of smoke hazard from fires in large spaces/discussion. *Am. Soc. Heat., Refrig. Air-Cond. Eng., Trans.* 106, 627.
- Nguyen, M.H., Ho, T.V., Zucker, J.-D., 2013. Integration of smoke effect and blind evacuation strategy (SEBES) within fire evacuation simulation. *Simul. Model. Pract. Theory* 36, 44–59.
- Peng, Y., Li, S., Hu, Z., 2019. A self-learning dynamic path planning method for evacuation in large public buildings based on neural networks. *Neurocomputing* 365, 71–85.
- Persson, S.M., Sharf, I., 2014. Sampling-based A* algorithm for robot path-planning. *Int. J. Robot. Res.* 33, 1683–1708.
- Pu, S., Zlatanova, S., 2005. Evacuation route calculation of inner buildings. *Geo-Information Disaster Manage.* 1143–1161.
- Ran, H., Sun, L., Gao, X., 2014. Influences of intelligent evacuation guidance system on crowd evacuation in building fire. *Autom. Constr.* 41, 78–82.
- Samah, K.A.F.A., Hussin, B., Basari, A.S.H., 2015. Modification of Dijkstra's algorithm for safest and shortest path during emergency evacuation. *Appl. Math. Sci.* 9, 1531–1541.
- Seike, M., Kawabata, N., Hasegawa, M., 2016. Experiments of evacuation speed in smoke-filled tunnel. *Tunn. Undergr. Space Technol.* 53, 61–67.
- Seike, M., Lu, Y.C., Kawabata, N., Hasegawa, M., 2021. Emergency evacuation speed distributions in smoke-filled tunnels. *Tunn. Undergr. Space Technol.* 112.
- Shen, Y., Wang, Q., Yan, W., Sun, J., 2015. An evacuation model coupling with toxic effect for chemical industrial park. *J. Loss Prev. Process Ind.* 33, 258–265.
- Shi, J., Ren, A., Chen, C., 2009. Agent-based evacuation model of large public buildings under fire conditions. *Autom. Constr.* 18, 338–347.
- Soltani, A.R., Tawfik, H., Goulermas, J.Y., Fernando, T., 2002. Path planning in construction sites: performance evaluation of the Dijkstra, A*, and GA search algorithms. *Adv. Eng. Inform.* 16, 291–303.
- Song, Y., Gong, J., Li, Y., Cui, T., Fang, L., Cao, W., 2013. Crowd evacuation simulation for bioterrorism in micro-spatial environments based on virtual geographic environments. *Saf. Sci.* 53, 105–113.
- Wang, Y., Zheng, X., 2013. Location optimization for evacuation signs and cellular automaton model simulation for evacuation in smoke. *J. Appl. Fire Sci.* 23, 283–297.
- Wong, S., Tang, P., Li, F., Wang, Z., Yu, S., 2015. Guidance path scheduling using particle swarm optimization in crowd simulation. *Comput. Animat. Virtual Worlds* 26, 387–395.
- Xu, L., Huang, K., Liu, J., Li, D., Chen, Y.F., 2022. Intelligent planning of fire evacuation routes using an improved ant colony optimization algorithm. *J. Build. Eng.* 61.
- Xu, M., Liu, Y., Huang, Q., Zhang, Y., Luan, G., 2007. An improved Dijkstra's shortest path algorithm for sparse network. *Appl. Math. Comput.* 185, 247–254.
- Yan, X., Yang, H., Mo, H., Xie, Y., Jin, Z., Zhou, Y., 2023. Numerical simulation on the smoke prevention performance of air curtains in an island-type subway station. *Fire* 6, 177.
- Zeng, W., Church, R.L., 2009. Finding shortest paths on real road networks: the case for A*. *Int. J. Geogr. Inf. Sci.* 23, 531–543.
- Zhang, P., Yang, L., Lo, S.M., Wang, D., Li, M., Jiang, J., Jiang, N., 2022a. Experimental study on evacuation behavior with guidance under high and low urgency conditions. *Saf. Sci.* 154.
- Zhang, Y., Li, W., Rui, Y., Wang, S., Zhu, H., Yan, Z., 2022b. A modified cellular automaton model of pedestrian evacuation in a tunnel fire. *Tunn. Undergr. Space Technol.* 130.
- Zhang, L., Liu, J., Tan, S., 2019. Multi-exit Evacuation strategy based on particle swarm optimization with background field. *J. Northeast. Univ. Nat. Sci.* 40, 1222–1227.
- Zhao, X., Ding, S., An, Y., Jia, W., 2018. Asynchronous reinforcement learning algorithms for solving discrete space path planning problems. *Appl. Intell.* 48, 4889–4904.
- Zhou, R., Cui, Y., Wang, Y., Jiang, J., 2021. A modified social force model with different categories of pedestrians for subway station evacuation. *Tunn. Undergr. Space Technol.* 110.